

MELANOMA

A Comprehensive Guide to Symptoms, Diagnosis, and Treatment

1. What is Melanoma?

Melanoma is a serious form of skin cancer that develops in melanocytes, the cells that produce melanin (the pigment that gives skin its color). While less common than other skin cancers, melanoma is more dangerous because it is more likely to spread (metastasize) to other parts of the body if not detected and treated early.

Key Facts about Melanoma:

- It is the most dangerous form of skin cancer
 - Can develop anywhere on the body, including areas not exposed to sun
 - Early detection significantly improves survival rates
 - Can occur in people of all skin types and ages
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2. Symptoms of Melanoma

The most important sign of melanoma is a change in an existing mole or the appearance of a new, unusual growth. The **ABCDE rule** is a helpful guideline for recognizing warning signs:

Letter	Sign	Description
A	Asymmetry	One half of the mole does not match the other half
B	Border	Irregular, scalloped, or poorly defined borders
C	Color	Varied shades of black, brown, tan, white, red, or blue
D	Diameter	Larger than 6mm (about the size of a pencil eraser)
E	Evolving	Changing in size, shape, color, or developing new symptoms

Other warning signs that may indicate melanoma include:

- A sore that does not heal
- A new spot or growth on the skin
- A mole that is raised or gets bigger
- Itching, tenderness, or bleeding from a mole
- Spreading of pigment from the border of a spot
- Redness or swelling beyond the mole

3. Risk Factors for Melanoma

Several factors can increase your risk of developing melanoma:

- **Excessive UV Exposure** – Sunburn, sunbathing, and tanning beds
 - **Fair Skin** – Less melanin provides less protection
 - **Family History** – Having a close relative with melanoma
 - **Many Moles** – Having more than 50 ordinary moles
 - **Age** – Risk increases with age
 - **Weak Immune System** – Due to disease or immunosuppressants
 - **Personal History** – Having had melanoma or other skin cancers before
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4. Diagnosis

If melanoma is suspected, a dermatologist will typically:

1. **Skin Examination** – Full body examination using a dermatoscope
 2. **Biopsy** – Taking a sample of the suspicious lesion for pathological analysis
 3. **Imaging Tests** – CT, MRI, or PET scans to check for spread
 4. **Sentinel Lymph Node Biopsy** – To check if cancer has spread
 5. **Genetic Testing** – To look for specific mutations (e.g., BRAF)
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5. Treatment Options

Treatment depends on the stage and location of the melanoma:

Treatment	Description	Stage
Surgery	Excision of the melanoma and surrounding skin	Early Stage
Immunotherapy	Drugs that help the immune system fight cancer	Advanced
Targeted Therapy	Drugs that target specific genetic mutations	Advanced
Radiation	High-energy radiation to kill cancer cells	Localized
Chemotherapy	Drugs to kill cancer cells throughout the body	Metastatic
Clinical Trials	New investigational treatments	All Stages

6. Prevention

- Use sunscreen with SPF 30+ and reapply every 2 hours
 - Seek shade during peak sun hours (10 am – 4 pm)
 - Wear protective clothing, including hats and sunglasses
 - Avoid tanning beds and sunlamps
 - Perform regular skin self-examinations
 - Visit a dermatologist annually for skin checks
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7. Survival Rates

Melanoma is highly treatable when caught early. Stage I survival rates exceed 95%. However, treatment becomes increasingly difficult if the disease spreads.

Stage	5-Year Survival Rate	Description
Stage 0	99%	In situ (confined to outer skin layer)
Stage I	90-95%	Localized, thin melanoma (< 2mm)
Stage II	75-80%	Localized, thick melanoma (> 2mm)
Stage III	50-70%	Spread to nearby lymph nodes
Stage IV	15-20%	Spread to distant organs

8. When to See a Doctor

You should see a doctor immediately if you notice:

- Any new mole or growth that looks different from other moles
 - Any mole that changes in size, shape, or color
 - A mole that itches, bleeds, or becomes painful
 - A sore that does not heal
 - Dark spots under the nails or on the palms/soles
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Sample Questions You Can Ask in Your RAG System

1. What are the ABCDE signs of melanoma?
2. How is melanoma diagnosed?
3. What are the survival rates for melanoma?
4. What treatment options are available for melanoma?
5. How can melanoma be prevented?
6. What are the risk factors for melanoma?
7. When should I see a doctor about a suspicious mole?

8. Is melanoma hereditary?
9. What is the difference between melanoma and other skin cancers?
10. How does melanoma spread to other parts of the body?